Data centre campus

Energy strategy report

Eemshaven 750 MW IT campus (Oostpolder)

Item	Value
Date of report	2026-10-31
Location	53.425, 6.875
Analysis start	2027-01
Currency shown	Euro
IT load at full build	750 MW in 6 phases
Options screened	2,000 of 52,023 combinations
Engine and app version	0.3.0
Inputs fingerprint	e4839996a2e832f7

Read this first

Concept-stage screening only. Figures are indicative and must be verified with project-specific data and specialist advice before any design or investment decision. This tool is decision support, not financial or engineering advice.

Every default is labelled "indicative — unverified" until it is researched. The tool makes no network connection; this report was made on this computer from the run shown. The Site tab (Beta) is not part of this report.

Summary

Written by fixed templates from this run's figures, without a language model: it describes the result under the run's weights and settings and does not advise.

Leading option

The option that leads under the current weights (cost 35 %, carbon 25 %, time-to-power 25 %, deliverability 15 %) is #1, Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · retained-plant backup · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 %, with a score of 89.3.

Why it leads

It is 0.09 points ahead of #2: 0.09 points from cost (LCOE €159 against €160/MWh).

What it gives up

On deliverability it has 89 points; the best of the adequate options, #415, has 96.8 points.

Confidence

11 other adequate options score within 1 point of it. In 300 Monte Carlo samples (top 20 options) it leads in 49 % and is in the top three in 94 %; it is also the robust pick (most often within 1 % of the best).

What would change the answer

#11 would lead if the deliverability weight fell from 15 % to 1 %, the other weights scaled in proportion. Breakevens: #4 would lead if IT load fell to × 0.978; #6 would lead if IT load fell to × 0.966.

Open flags on the leader

Resilience: meets the requirement with flags (backup depends on gas network availability during a grid outage). Permitting: aggregated combustion thermal input 2415 MWth >= 50 MWth (IED threshold), a major flag, and 1 minor flag.

Inputs with most influence

Of the 7 inputs tested on #1, grid import price (confidence C) moves LCOE by up to €24.1/MWh across its range.

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Summary detail

Leading option

- LCOE €159/MWh, capex €1.17 bn, carbon 175 g/kWh (location-based), power on time.
- In 300 Monte Carlo samples its LCOE is €151 to €182/MWh (P10 to P90; P50 €167/MWh).
- Score 89.26; axis scores: cost 90.8, carbon 80.0, time-to-power 100.0, deliverability 83.2.
- NPV of net cost €6.79 bn; hourly carbon-free energy 58 %; deliverability 89 points.
- Monte Carlo P10–P90: NPV of net cost €5.45 bn to €8.12 bn; capex €1.02 bn to €1.56 bn; carbon 169 to 202 g/kWh (location-based).
- Tornado (one input at a time): LCOE €149 to €174/MWh.

Why it leads

- #2 differs in 5 choices, among them backup power technology: Diesel gensets.
- They are level on carbon, time-to-power and deliverability.
- #2: Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · Diesel N+1 · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 % (score 89.17).
- Choices of #2 that differ from #1: C5 Backup power technology: Diesel gensets; C5a Backup arrangement: Per building; C6 Backup topology: N+1; C7 Autonomy / fuel storage: 48 h; C18 Genset arrangement: Single-deck.
- Cost: LCOE €159 against €160/MWh; weighted difference +0.09 points, in favour of #1.
- Carbon: 175 against 175.1 g/kWh; no weighted difference (less than 0.01 points).
- Time-to-power: both on time; no weighted difference (less than 0.01 points).
- Deliverability: both 89 points; no weighted difference (less than 0.01 points).

What it gives up

- On carbon it has 175 g/kWh; the best of the adequate options, #956, has 164 g/kWh.
- Cost: €159/MWh; best of the adequate options: #166 Grid + gas recips bridge -> keep shortfall share as prime, rest as backup · HVO N+1 pooled · Evaporative-adiabatic · Wind 10 MW · PPA hybrid 100 %, €157/MWh (axis score 91.4 against 90.8).
- Carbon: 175 g/kWh; best of the adequate options: #956 Grid + fuel cells bridge -> retain as backup · Gas CHP-trigen top-up · Diesel N+1

· Evaporative-adiabatic · PV 42 MWp · Wind 10 MW · BESS 4 h · PPA hybrid 100 %, 164 g/kWh (axis score 81.7 against 80.0).

- Deliverability: 89 points; best of the adequate options: #415 Grid + fuel cells bridge -> keep shortfall share as prime, rest as backup · retained-plant backup · Evaporative-adiabatic · PV 42 MWp · BESS 4 h · PPA hybrid 100 %, 96.8 points (axis score 100.0 against 83.2).
- Outside its group: #1998 Grid · SMR · HVO N+1 · Evaporative-adiabatic does better on carbon (52.4 g/kWh against 175 g/kWh).
- Outside its group: #1642 Grid + fuel cells bridge -> retain as backup · retained-plant backup · Evaporative-adiabatic · PV 33 MWp · BESS for the shortfall does better on deliverability (96.3 points against 89 points).
- Outside its group: #1998 Grid · SMR · HVO N+1 · Evaporative-adiabatic does better on cost (€135/MWh against €159/MWh).

Confidence

- It uses 12 key inputs left at their defaults; 550 of 574 default records are indicative — unverified.
- The Monte Carlo ran on representative days: adequacy and the order of the options are approximate, and now and then a full-year run would put a different option first in a sample.
- Key inputs at their defaults: Analysis start; Average IT utilisation; Ramp; Load shape; Grid non-commodity adder; Electricity price escalation; Gas commodity price; Carbon price (Scope 1); WACC (real); Evaluation period; UPS ride-through; PPA type.
- Other adequate options close to the leader: 1 within 0.1 points, 11 within 1 point, 222 within 5 points.

What would change the answer

- #312 would lead if the cost weight fell from 35 % to 13 %.
- The deliverability weight fell from 15 % to 1 % (crossing at 1.1 %): #11 Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · HVO N+1 pooled · Evaporative-adiabatic · PV 42 MWp · PPA hybrid 100 % leads.
- The cost weight fell from 35 % to 13 % (crossing at 13.7 %): #312 Grid + fuel cells bridge -> keep shortfall share as prime, rest as backup · retained-plant backup · Evaporative-adiabatic · PV 42 MWp · PPA hybrid 100 % leads.
- The carbon weight fell from 25 % to 1 % (crossing at 1.1 %): #4 Grid + gas recips bridge -> keep shortfall share as prime, rest as backup · retained-plant backup · Evaporative-adiabatic · BESS for the shortfall · PPA hybrid 100 % leads.
- The deliverability weight rose from 15 % to 40 % (crossing at 39.8 %): #312 Grid + fuel cells bridge -> keep shortfall share as prime, rest as backup · retained-plant backup · Evaporative-adiabatic · PV 42 MWp · PPA hybrid 100 % leads.
- The cost weight rose from 35 % to 86 % (crossing at 85.1 %): #30 Grid + gas recips bridge -> keep shortfall share as prime, rest as backup · HVO N+1 pooled · Evaporative-adiabatic · PV 42 MWp · PPA hybrid 100 % leads.
- The carbon weight rose from 25 % to 87 % (crossing at 86.4 %): #312 Grid + fuel cells bridge -> keep shortfall share as prime, rest as backup · retained-plant backup · Evaporative-adiabatic · PV 42 MWp · PPA hybrid 100 % leads.
- Not listed: a weight at exactly 100 %, where many options tie on that axis and the order between them is by option code, and lowering a weight that is already 100 %.
- #4 would lead if IT load fell to × 0.978 (5 % of the driver's range from today's value).
- #6 would lead if IT load fell to × 0.966 (9 % of the driver's range from today's value).
- #2 would lead if the grid full-connection date slipped by 9 months or more (30 % of the driver's range from today's value).
- The breakeven search checks a grid of points on each side of today's value: a swap that reverses again between two checked points is not seen.

Inputs with most influence

- Next: WACC (C), €22.3/MWh; capex (C), €19.4/MWh.
- Ordered by the swing in LCOE; data confidence breaks ties (C, the least certain, first).
- Grid import price (indicative — unverified, confidence C): LCOE €149/MWh at × 0.7, €174/MWh at × 1.5.
- WACC (real) (indicative — unverified, confidence C): LCOE €152/MWh at 4 %, €174/MWh at 12 %.
- Capex (every line) (indicative — unverified, confidence C): LCOE €152/MWh at × 0.7, €171/MWh at × 1.5.
- Gas commodity price (indicative — unverified, confidence C): LCOE €154/MWh at × 0.7, €164/MWh at × 1.3.
- Grid full-connection date (indicative — unverified, confidence C): LCOE €157/MWh at -6 months, €165/MWh at +24 months.
- Carbon price (Scope 1) (indicative — unverified, confidence C): LCOE €155/MWh at × 0.5, €163/MWh at × 1.5.
- Capacity factors (on-site solar and wind) (indicative — unverified, confidence C): LCOE €159/MWh at × 0.85, €159/MWh at × 1.15.

The flags of the leader are listed in full in section 8.

1. Inputs

Every input of the run, with its value and whether you entered it or the tool used its default. A default is labelled with the registry label and its basis (source, judgement or run setting). Site geometry (boundary, exclusions, height zones, connection points, receptors) belongs to the Site tab (Beta) and is not listed.

Project

Input	Value	Entered or default
project.name	Eemshaven 750 MW IT campus (Oostpolder)	Entered
project.lat	53.425	Entered (required)
project.lon	6.875	Entered (required)
project.country	NL	Entered
project.currency	by country	Default: indicative — unverified (judgement)
project.analysisStart	2027-01	Default: indicative — unverified (judgement)
project.seed	20261015 integer	Default: indicative — unverified (run setting)

Site (area and land)

Input	Value	Entered or default
site.areaHa	120	Entered (required)
site.aspect	1.5 ratio	Default: indicative — unverified (judgement)
site.setbackM	10 m	Default: indicative — unverified (judgement)
site.heightLimitM	25 m	Default: indicative — unverified (judgement)
site.softUsableBySolar	yes	Default: indicative — unverified (judgement)
site.roadsFraction	0.15 fraction of net developable area	Default: indicative — unverified (judgement)
site.pvRoofM2	not set	Default: indicative — unverified (judgement)
site.pvGroundHa	0 ha of ground available for PV	Default: indicative — unverified (judgement)
site.offsiteLandHa	0 ha	Default: indicative — unverified (judgement)
site.offsiteDistanceKm	2 km private-wire distance	Default: indicative — unverified (judgement)
site.groundConditions	assumed suitable	Default: indicative — unverified (judgement)
site.areaFromBoundary	no	Entered

Load

Input	Value	Entered or default
load.itMW	750 MW IT at full build (blank = estimated from the site area)	Entered
load.estimateCoverage	0.35 fraction of the site area covered by data halls	Default: indicative — unverified (judgement)
load.phases	1: itMW 125, date 2029-01; 2: itMW 125, date 2029-10; 3: itMW 125, date 2030-07; 4: itMW 125, date 2031-04; 5: itMW 125, date 2032-01; 6: itMW 125, date 2032-10	Entered
load.rampStart	0.25 fraction of phase IT at energisation	Default: indicative — unverified (judgement)
load.rampMonths	12 months	Default: indicative — unverified (judgement)
load.utilisation	0.8 fraction of installed IT	Default: indicative — unverified (judgement)
load.workload	cloud	Default: indicative — unverified (judgement)
load.shape	daily	Default: indicative — unverified (judgement)
load.dailySwing	not set	Default: indicative — unverified (judgement)
load.aiTrainUtil	0.9 average fraction of installed IT	Default: indicative — unverified (judgement)
load.aiTrainDip	0.55 fraction of installed IT during a checkpoint dip	Default: indicative — unverified (judgement)

Input	Value	Entered or default
load.aiTrainDipEveryH	6 h between checkpoint dips	Default: indicative — unverified (judgement)
load.aiTrainDipH	1 h per dip	Default: indicative — unverified (judgement)
load.aiInfUtil	0.7 average fraction of installed IT	Default: indicative — unverified (judgement)
load.aiInfTrough	0.6 night trough as a fraction of the daily peak	Default: indicative — unverified (judgement)
load.aiInfPeakHour	15 hour of the daily peak (local)	Default: indicative — unverified (judgement)
load.aiInfWeekend	0.9 weekend level as a fraction of weekdays	Default: indicative — unverified (judgement)
load.coolingApproach	evap	Entered
load.grossUp	2.2 GFA / whitespace	Default: indicative — unverified (judgement)
load.storeys	2 storeys	Default: indicative — unverified (judgement)
load.storeyHeightM	6 m	Default: indicative — unverified (judgement)
load.roofZoneM	4 m	Default: indicative — unverified (judgement)
load.coolingFactor	1 × cooling energy (PUE above 1)	Default: indicative — unverified (judgement)
load.ancillaryFrac	0.02 fraction of installed IT	Default: indicative — unverified (judgement)
load.flexMaxPct	0.1 fraction of IT curtailable	Default: indicative — unverified (judgement)
load.flexHours	100 h/yr	Default: indicative — unverified (judgement)
load.flexTrigger	price	Default: indicative — unverified (judgement)
load.flexTriggerPrice	250 €/MWh grid price (wholesale × multiplier + adder)	Default: indicative — unverified (judgement)
load.flexTriggerCarbon	400 gCO2e/kWh grid carbon intensity	Default: indicative — unverified (judgement)
load.flexTriggerTemp	28 °C ambient dry-bulb (weather and grid stress)	Default: indicative — unverified (judgement)
load.flexEventH	4 h per event (maximum)	Default: indicative — unverified (judgement)
load.flexNoticeH	1 h notice	Default: indicative — unverified (judgement)
load.flexRecoveryH	12 h between events	Default: indicative — unverified (judgement)
load.flexCostPerMWh	500 €/MWh of IT load curtailed	Default: indicative — unverified (judgement)
load.flexBatteryFixed	threshold	Default: indicative — unverified (judgement)
load.flexBatteryHours	100 h/yr of trigger events covered (when hours are held fixed)	Default: indicative — unverified (judgement)
load.flexBatteryMW	not set	Default: indicative — unverified (judgement)
load.flexBatteryMWh	not set	Default: indicative — unverified (judgement)
load.flexBatteryResponse	yes	Default: indicative — unverified (judgement)

Grid

Input	Value	Entered or default
grid.nowMW	0 MW import available now (blank = not known: 0 MW)	Entered
grid.fullMW	950 MW import at full connection (blank = not known)	Entered
grid.fullDate	2033-01	Entered
grid.unknownNowMW	0 MW now when not known	Default: indicative — unverified (judgement)
grid.unknownFullPerITMW	1.4 MW of grid per MW of IT when not known	Default: indicative — unverified (judgement)
grid.unknownFullMonths	48 months from the analysis start when not known	Default: indicative — unverified (judgement)
grid.steps	none	Default: indicative — unverified (judgement)
grid.exportMW	0 MW	Default: indicative — unverified (judgement)
grid.exportFactor	0.9 × hourly price	Default: indicative — unverified (judgement)
grid.priceAdder	by country	Default: indicative — unverified (source)
grid.priceMultiplier	1 × country profile	Default: indicative — unverified (judgement)
grid.carbonMultiplier	1 × country profile	Default: indicative — unverified (judgement)
grid.contractWaitMW	not set	Default: indicative — unverified (judgement)
grid.contractBridgeMW	not set	Default: indicative — unverified (judgement)

Input	Value	Entered or default
grid.contractExportMW	not set	Default: indicative — unverified (judgement)
grid.connectionCharge	0 € (lump sum, the same for every grid-connected option)	Default: indicative — unverified (judgement)
grid.dateUncertainty	-6; 24	Default: indicative — unverified (judgement)

Utilities and PPA

Input	Value	Entered or default
utilities.gasAvailable	Yes	Entered
utilities.gasDateMonths	0 months from project start	Entered
utilities.gasPrice	by country	Default: indicative — unverified (source)
utilities.hydrogenAvailable	No	Default: indicative — unverified (judgement)
utilities.waterAvailable	Unknown	Default: indicative — unverified (judgement)
utilities.districtHeating	No	Default: indicative — unverified (judgement)
utilities.reYieldFactor	1 × on-site solar and wind output (capacity factors)	Default: indicative — unverified (judgement)
utilities.ppaInclude	yes	Entered
utilities.ppaType	hybrid	Default: indicative — unverified (judgement)
utilities.ppaCoverage	100 % of each year's facility consumption	Default: indicative — unverified (judgement)

Existing plant near the site (lease)

Input	Value	Entered or default
existing.include	no	Default: indicative — unverified (judgement)
existing.name	Existing plant	Default: indicative — unverified (judgement)
existing.type	ccgt	Default: indicative — unverified (judgement)
existing.units	not set	Default: indicative — unverified (judgement)
existing.unitMW	not set	Default: indicative — unverified (judgement)
existing.availableDate	not set	Default: indicative — unverified (judgement)
existing.eff	not set	Default: indicative — unverified (judgement)
existing.msl	not set	Default: indicative — unverified (judgement)
existing.vomPerMWh	not set	Default: indicative — unverified (judgement)
existing.leasePerKWMonth	6 €/kW/month	Default: indicative — unverified (judgement)
existing.linkCapex	0 €	Default: indicative — unverified (judgement)

Requirements

Input	Value	Entered or default
requirements.backedPct	1 fraction of IT + essential mechanical	Default: indicative — unverified (judgement)
requirements.topology	N+1	Default: indicative — unverified (judgement)
requirements.autonomyH	48 h	Default: indicative — unverified (judgement)
requirements.islanding	No	Default: indicative — unverified (judgement)
requirements.upsMin	5 min at full IT	Default: indicative — unverified (judgement)
requirements.blockSize	10 units per block	Default: indicative — unverified (judgement)
requirements.catcherN	4 blocks per catcher unit	Default: indicative — unverified (judgement)
requirements.carbonTarget	not set	Default: indicative — unverified (judgement)
requirements.capexCeiling	not set	Default: indicative — unverified (judgement)
requirements.maxRunHours	not set	Default: indicative — unverified (judgement)
requirements.iedMWth	50 MWth aggregated	Default: indicative — unverified (source)
requirements.mcpMinMWth	1 MWth per unit	Default: indicative — unverified (source)

Input	Value	Entered or default
requirements.mcpMaxMWth	50 MWth per unit	Default: indicative — unverified (source)
requirements.sevesoH2t	5 t hydrogen stored	Default: indicative — unverified (source)
requirements.permitAllowanceM	6 months before ordering plant that needs significant environmental permitting (IED, Seveso)	Default: indicative — unverified (judgement)
requirements.sevesoFuelT	2500 t diesel/HVO stored	Default: indicative — unverified (source)
requirements.testRunHours	6 h/yr per unit	Default: indicative — unverified (judgement)
requirements.gensetPrimaryHours	0 h/yr	Default: indicative — unverified (judgement)
requirements.testRunLoad	0.75 fraction of rating	Default: indicative — unverified (judgement)
requirements.bessUpsGensetFraction	1 genset capacity as fraction of critical load at topology N	Default: indicative — unverified (judgement)
requirements.landTakeCeilingHa	not set	Default: indicative — unverified (judgement)
requirements.noiseNightDbA	45 dB(A) at receptors, night	Default: indicative — unverified (judgement)

Economics

Input	Value	Entered or default
economics.periodYears	20 yr	Default: indicative — unverified (judgement)
economics.wacc	0.07 real	Default: indicative — unverified (judgement)
economics.contingency	0.1 fraction of capex	Default: indicative — unverified (judgement)
economics.capexLow	-0.3 fraction (AACE Class 5)	Default: indicative — unverified (source)
economics.capexHigh	0.5 fraction (AACE Class 5)	Default: indicative — unverified (source)
economics.carbonPrice	by country	Default: indicative — unverified (source)
economics.carbonEscalation	0.02 real per yr	Default: indicative — unverified (source)
economics.capexFactor	1 × every capex line	Default: indicative — unverified (judgement)
economics.bessCapexFactor	1 × battery capex	Default: indicative — unverified (judgement)
economics.gridCapexFactor	1 × grid connection capex	Default: indicative — unverified (judgement)
economics.elecEscalation	0 real per yr	Default: indicative — unverified (source)
economics.gasEscalation	0 real per yr	Default: indicative — unverified (source)
economics.salvageFraction	0.5 of remaining book value when removed	Default: indicative — unverified (judgement)
economics.revenueExport	no	Default: indicative — unverified (judgement)
economics.displayPeriods	10; 15	Default: indicative — unverified (judgement)

Weights

Input	Value	Entered or default
weights.cost	35 raw 0–100	Default: indicative — unverified (judgement)
weights.carbon	25 raw 0–100	Default: indicative — unverified (judgement)
weights.ttp	25 raw 0–100	Default: indicative — unverified (judgement)
weights.deliverability	15 raw 0–100	Default: indicative — unverified (judgement)

Deliverability

Input	Value	Entered or default
deliverability.emergingPenalty	15 points per 100 % share	Default: indicative — unverified (judgement)
deliverability.nascentPenalty	40 points per 100 % share	Default: indicative — unverified (judgement)
deliverability.majorFlagPenalty	8 points per major permitting flag	Default: indicative — unverified (judgement)
deliverability.minorFlagPenalty	3 points per minor permitting flag	Default: indicative — unverified (judgement)
deliverability.gasUnknownPenalty	10 points	Default: indicative — unverified (judgement)
deliverability.hydrogenPenalty	10 points	Default: indicative — unverified (judgement)

Input	Value	Entered or default
deliverability.leadTimePenalty	5 points per technology	Default: indicative — unverified (judgement)
deliverability.leadTimeThresholdM	24 months procurement	Default: indicative — unverified (judgement)
deliverability.pooledBackupRisk	minor	Default: indicative — unverified (judgement)

Run settings

Input	Value	Entered or default
settings.generationCap	2000 options	Default: indicative — unverified (run setting)
settings.sampleAboveCap	yes	Default: indicative — unverified (judgement)
settings.gridHandling	unmet	Default: indicative — unverified (judgement)
settings.runHourHandling	flag	Default: indicative — unverified (judgement)
settings.costMetric	lcoe	Default: indicative — unverified (judgement)
settings.carbonMetric	location	Default: indicative — unverified (judgement)
settings.absoluteScaling	no	Default: indicative — unverified (judgement)
settings.rankCompliantFirst	no	Default: indicative — unverified (judgement)
settings.hideDominated	no	Default: indicative — unverified (judgement)
settings.windSiteThresholdHa	20 ha	Default: indicative — unverified (judgement)
settings.bridgeBessHours	24 h	Default: indicative — unverified (judgement)
settings.timelineMonths	120 months	Default: indicative — unverified (run setting)
settings.rentalLeadMonths	1 months	Default: indicative — unverified (judgement)
settings.maxDispatchYears	15 years	Default: indicative — unverified (run setting)
settings.screeningModel	full	Default: indicative — unverified (run setting)
settings.repDaysPerMonth	3 typical days a month (plus the dearest day and the design day)	Default: indicative — unverified (run setting)
settings.verifyTop	10 options verified with the full year	Default: indicative — unverified (run setting)
settings.autonomyBasis	critical load	Default: indicative — unverified (judgement)
settings.layoutCellM	2 m raster cell	Default: indicative — unverified (run setting)
settings.layoutCellLargeM	5 m raster cell (large sites)	Default: indicative — unverified (run setting)
settings.siteLayoutScope	auto	Default: indicative — unverified (run setting)
settings.siteLayoutWorkLimit	20000000 block2 (ground blocks squared, summed over the options)	Default: indicative — unverified (judgement)
settings.layoutLargeSiteHa	50 ha gross site area above which the large cell is used	Default: indicative — unverified (run setting)

Technology choices generated

Choice	State	Values generated
C1 Grid strategy	open	Wait for grid, Bridge until grid, Islanded
C2 Bridging supply	open	Gas recips, Gas turbines, Fuel cells, HVO-diesel prime, Rented temporary generation
C3 Bridging plant after grid arrives	open	Remove, Retain as backup, Retain as prime, Keep shortfall share as prime, rest as backup
C4 Primary firm on-site generation	open	None, Gas CHP-trigen, Fuel cells
C4s Firm generation sizing	open	Average load (baseload), Permanent shortfall (top-up)
C5 Backup power technology	open	HVO gensets, Diesel gensets, Retained bridging plant
C5a Backup arrangement	open	Per building, Pooled across the campus
C6 Backup topology	locked	N+1
C7 Autonomy / fuel storage	locked	48 h
C8 Cooling approach	locked	Evaporative-adiabatic

Choice	State	Values generated
C9 Heat rejection location	open	Roof, Ground, Split
C10 On-site solar	open	None, Roof only
C11 On-site wind	open	None, Fit as many as separations allow, Off-site land
C12 BESS size	open	None, 2 h at 25 %, 4 h at 50 %, Sized to the permanent shortfall, Sized to the trigger hours
C12r BESS role	locked	Peak-shave against grid cap
C13 Offsite PPA	open	None, Hybrid
C14 PPA coverage	locked	100 %
C15 Heat reuse	locked	None
C16 Demand flexibility	locked	Off
C17 Substation type	open	AIS
C18 Genset arrangement	open	Single-deck
C19 Phasing of energy plant	locked	Build per phase
C20 Fuel decarbonisation pathway	locked	None

2. Assumptions

154 inputs use their default. The registry holds 574 records (inputs, technology, cost, carbon, spatial and run settings); 550 are "indicative — unverified" and 24 are "sourced — not independently checked" (researched, with source, year and link, approved by the owner, not checked by an independent person). None is presented as verified. The full registry is in the user guide's appendix, and the run log lists the records each run read.

Default used	Value	Basis and note
project.currency	by country	indicative — unverified (judgement). Display only; converted at the ECB reference rates of the release (project.fx). Inputs stay in euro.
project.analysisStart	2027-01	indicative — unverified (judgement). Today + 3 months. No source; replace with the client programme start.
project.seed	20261015 integer	indicative — unverified (run setting)
site.aspect	1.5 ratio	indicative — unverified (judgement). No source.
site.setbackM	10 m	indicative — unverified (judgement). No source; replace with local planning requirement.
site.heightLimitM	25 m	indicative — unverified (judgement)
site.softUsableBySolar	yes	indicative — unverified (judgement). Soft exclusions can be used by ground solar (configurable). Off: ground solar does not use them either. Other blocks never use a soft exclusion by default; only at the last relaxation level (L4), flagged "Block in soft exclusion" and "Fits with flags" (§10.4 step 5, §15).
site.roadsFraction	0.15 fraction of net developable area	indicative — unverified (judgement). Applied to the buildable envelope area: allowance = fraction × envelope, net developable = envelope - allowance (the brief's definition is circular). Held back as a land budget, not a located area.
site.pvRoofM2	not set	indicative — unverified (judgement). OPEN-QUESTIONS 53 (2026-10-22): blank = today's rule, 50 % of the free roof after roof plant is usable (solarRoof.usableFraction). An entered area replaces the rule; car-park canopies can be included here or in the ground area.
site.pvGroundHa	0 ha of ground available for PV	indicative — unverified (judgement). OPEN-QUESTIONS 53 (2026-10-22): 0 = no ground solar.
site.offsiteLandHa	0 ha	indicative — unverified (judgement)
site.offsiteDistanceKm	2 km private-wire distance	indicative — unverified (judgement). Used only when off-site land or off-site plant exists; replace with the real route length.
site.groundConditions	assumed suitable	indicative — unverified (judgement). Note only; no geotechnical modelling (flat site, §10.1).
load.estimateCoverage	0.35 fraction of the site area covered by data halls	indicative — unverified (judgement). Walkthrough 2026-10-21: 35 % site coverage for the IT estimate from area. To research.
load.rampStart	0.25 fraction of phase IT at energisation	indicative — unverified (judgement). The owner's judgement (2026-10-22, OPEN-QUESTIONS 51; was 50 %). No source; replace with the client's fit-out programme.

Default used	Value	Basis and note
load.rampMonths	12 months	indicative — unverified (judgement)
load.utilisation	0.8 fraction of installed IT	indicative — unverified (judgement)
load.workload	cloud	indicative — unverified (judgement). Mixed cloud
load.shape	daily	indicative — unverified (judgement). Daily fluctuation is the shape built before work package 4.
load.dailySwing	not set	indicative — unverified (judgement). Blank = by workload, as built: ±1 % AI training, ±8 % enterprise, ±4 % otherwise.
load.aiTrainUtil	0.9 average fraction of installed IT	indicative — unverified (judgement). Mock-up value (walkthrough 2026-10-21); to research.
load.aiTrainDip	0.55 fraction of installed IT during a checkpoint dip	indicative — unverified (judgement). Mock-up value; to research.
load.aiTrainDipEveryH	6 h between checkpoint dips	indicative — unverified (judgement). Mock-up value; to research.
load.aiTrainDipH	1 h per dip	indicative — unverified (judgement). Mock-up value; to research. The hourly model cannot show dips shorter than an hour.
load.aiInfUtil	0.7 average fraction of installed IT	indicative — unverified (judgement). No source; to research.
load.aiInfTrough	0.6 night trough as a fraction of the daily peak	indicative — unverified (judgement). No source; to research.
load.aiInfPeakHour	15 hour of the daily peak (local)	indicative — unverified (judgement). No source; to research.
load.aiInfWeekend	0.9 weekend level as a fraction of weekdays	indicative — unverified (judgement). No source; to research.
load.grossUp	2.2 GFA / whitespace	indicative — unverified (judgement)
load.storeys	2 storeys	indicative — unverified (judgement)
load.storeyHeightM	6 m	indicative — unverified (judgement)
load.roofZoneM	4 m	indicative — unverified (judgement)
load.coolingFactor	1 × cooling energy (PUE above 1)	indicative — unverified (judgement). Tornado driver "PUE": scales fans and mechanical cooling. No source.
load.ancillaryFrac	0.02 fraction of installed IT	indicative — unverified (judgement)
load.flexMaxPct	0.1 fraction of IT curtailable	indicative — unverified (judgement)
load.flexHours	100 h/yr	indicative — unverified (judgement)
load.flexTrigger	price	indicative — unverified (judgement). Weather and grid stress is modelled as the ambient temperature at or above the trigger level (source to research).
load.flexTriggerPrice	250 €/MWh grid price (wholesale × multiplier + adder)	indicative — unverified (judgement). Mock-up value; to research against ENTSO-E price history.
load.flexTriggerCarbon	400 gCO2e/kWh grid carbon intensity	indicative — unverified (judgement). No source; to research.
load.flexTriggerTemp	28 °C ambient dry-bulb (weather and grid stress)	indicative — unverified (judgement). No source; to research.
load.flexEventH	4 h per event (maximum)	indicative — unverified (judgement). Mock-up value.
load.flexNoticeH	1 h notice	indicative — unverified (judgement). Mock-up value. Recorded, not modelled: the hourly model knows prices and carbon in advance (day-ahead), so notice does not change the hours.
load.flexRecoveryH	12 h between events	indicative — unverified (judgement). Mock-up value.
load.flexCostPerMWh	500 €/MWh of IT load curtailed	indicative — unverified (judgement). Mock-up value: the cost of curtailing customer IT; to research.
load.flexBatteryFixed	threshold	indicative — unverified (judgement). Walkthrough 2026-10-21: the engineer chooses which is held fixed. Threshold (default): the battery is sized to cover the trigger hours at the trigger level. Hours: the trigger level is set so the events reach the target hours, and the battery covers them. Size: the battery size is entered and the hours it covers are reported.
load.flexBatteryHours	100 h/yr of trigger events covered (when hours are held fixed)	indicative — unverified (judgement)
load.flexBatteryMW	not set	indicative — unverified (judgement). Blank = the trigger-hours size.
load.flexBatteryMWh	not set	indicative — unverified (judgement). Blank = the trigger-hours size.
load.flexBatteryResponse	yes	indicative — unverified (judgement). Walkthrough 2026-10-21: the economic battery runs the site in trigger hours before any curtailment.

Default used	Value	Basis and note
grid.unknownNowMW	0 MW now when not known	indicative — unverified (judgement). No source. Country values to research (package 0).
grid.unknownFullPerITMW	1.4 MW of grid per MW of IT when not known	indicative — unverified (judgement). About the design facility load (IT × design-day PUE + ancillary). No source; country values to research.
grid.unknownFullMonths	48 months from the analysis start when not known	indicative — unverified (judgement). No source; country connection lead times to research (package 0).
grid.steps	none	indicative — unverified (judgement)
grid.exportMW	0 MW	indicative — unverified (judgement)
grid.exportFactor	0.9 × hourly price	indicative — unverified (judgement)
grid.priceAdder	by country	indicative — unverified (source)
grid.priceMultiplier	1 × country profile	indicative — unverified (judgement)
grid.carbonMultiplier	1 × country profile	indicative — unverified (judgement)
grid.contractWaitMW	not set	indicative — unverified (judgement). OPEN-QUESTIONS 19: the contracted grid capacity of every C1 = "Wait for grid" option; the import limit is the lower of this and the offer in each period. Blank = the full offer (as built before).
grid.contractBridgeMW	not set	indicative — unverified (judgement). OPEN-QUESTIONS 19: the contracted grid capacity of every C1 = "Bridge until grid" option, for example 80 MW of a 120 MW offer. Blank = the full offer.
grid.contractExportMW	not set	indicative — unverified (judgement). OPEN-QUESTIONS 19: the contracted import capacity of C1 = "Grid + export" options. Blank = the full offer.
grid.connectionCharge	0 € (lump sum, the same for every grid-connected option)	indicative — unverified (judgement). Walkthrough 2026-10-21: default 0, the same for every option; OPEN-QUESTIONS 20: a lump sum, not paid by islanded options (0 MW contracted); OPEN-QUESTIONS 50: the expected cost (standard rate per MW × contracted MW) is shown next to the field. Every option relying on the default 0 is flagged "assumed".
grid.dateUncertainty	-6; 24	indicative — unverified (judgement). Range of the grid-date driver in the tornado, the breakevens and the Monte Carlo (work packages 9, 17, 18).
utilities.gasPrice	by country	indicative — unverified (source)
utilities.hydrogenAvailable	No	indicative — unverified (judgement)
utilities.waterAvailable	Unknown	indicative — unverified (judgement)
utilities.districtHeating	No	indicative — unverified (judgement)
utilities.reYieldFactor	1 × on-site solar and wind output (capacity factors)	indicative — unverified (judgement). Tornado driver "capacity factors". No source.
utilities.ppaType	hybrid	indicative — unverified (judgement)
utilities.ppaCoverage	100 % of each year's facility consumption	indicative — unverified (judgement). One coverage, set by the engineer (OPEN-QUESTIONS 49).
existing.include	no	indicative — unverified (judgement). Off unless the engineer enters an existing plant near the site (work package 12).
existing.name	Existing plant	indicative — unverified (judgement)
existing.type	ccgt	indicative — unverified (judgement). Sets the efficiency, minimum stable load and variable O&M unless entered.
existing.units	not set	indicative — unverified (judgement). No default: enter the units the owner offers. Used exactly, with no redundancy added.
existing.unitMW	not set	indicative — unverified (judgement). No default: net output per unit at site conditions.
existing.availableDate	not set	indicative — unverified (judgement). Empty = available from the analysis start (the plant exists). The month the lease and the connection to the campus can be ready.
existing.eff	not set	indicative — unverified (judgement). Empty = by type.
existing.msl	not set	indicative — unverified (judgement). Empty = by type.
existing.vomPerMWh	not set	indicative — unverified (judgement). Empty = by type.
existing.leasePerKWMonth	6 €/kW/month	indicative — unverified (judgement). Capacity payment for the lease, paid from a month before the plant is first needed (settings.rentalLeadMonths). Assumed in the Pego study (2026-10-24); replace with the owner's offer.
existing.linkCapex	0 €	indicative — unverified (judgement). One-off: the connection from the plant to the campus (lines, bays). Booked when the lease starts.
requirements.backedPct	1 fraction of IT + essential mechanical	indicative — unverified (judgement)
requirements.topology	N+1	indicative — unverified (judgement)

Default used	Value	Basis and note
requirements.autonomyH	48 h	indicative — unverified (judgement)
requirements.islanding	No	indicative — unverified (judgement). Default not stated in brief; "No" chosen (judgement).
requirements.upsMin	5 min at full IT	indicative — unverified (judgement). The resilience reserve battery (walkthrough 2026-10-21): always kept, never used for economic duties; longer autonomy can be chosen site-wide.
requirements.blockSize	10 units per block	indicative — unverified (judgement)
requirements.catcherN	4 blocks per catcher unit	indicative — unverified (judgement)
requirements.carbonTarget	not set	indicative — unverified (judgement). None
requirements.capexCeiling	not set	indicative — unverified (judgement). None
requirements.maxRunHours	not set	indicative — unverified (judgement). None
requirements.iedMWth	50 MWth aggregated	indicative — unverified (source)
requirements.mcpMinMWth	1 MWth per unit	indicative — unverified (source)
requirements.mcpMaxMWth	50 MWth per unit	indicative — unverified (source)
requirements.sevesoH2t	5 t hydrogen stored	indicative — unverified (source)
requirements.permitAllowanceM	6 months before ordering plant that needs significant environmental permitting (IED, Seveso)	indicative — unverified (judgement). Owner's decision 2026-10-22 (OPEN-QUESTIONS 44, 45): plant is ordered from day one; only plant under the IED or Seveso waits this allowance. Wind, ground solar and SMRs keep their own planning months (OPEN-QUESTIONS 56).
requirements.sevesoFuelT	2500 t diesel/HVO stored	indicative — unverified (source)
requirements.testRunHours	6 h/yr per unit	indicative — unverified (judgement). Walkthrough 2026-10-21: 6 h a year per generator (replaces 12 h), about monthly 30-minute exercise runs.
requirements.gensetPrimaryHours	0 h/yr	indicative — unverified (judgement). Walkthrough 2026-10-21: backup gensets can run as primary power for a set number of hours a year, dispatched and costed. Default 0: they never run in normal dispatch. Hours chosen: the most expensive grid-price hours of each year.
requirements.testRunLoad	0.75 fraction of rating	indicative — unverified (judgement). Walkthrough 2026-10-21: test load 75 %, a separate editable input.
requirements.bessUpsGensetFraction	1 genset capacity as fraction of critical load at topology N	indicative — unverified (judgement). BESS-augmented UPS replaces redundant genset units with BESS for its duration.
requirements.landTakeCeilingHa	not set	indicative — unverified (judgement). Flag only.
requirements.noiseNightDbA	45 dB(A) at receptors, night	indicative — unverified (judgement). No source; replace with the local planning or environmental noise criterion. Screening proxy, not an acoustic assessment.
economics.periodYears	20 yr	indicative — unverified (judgement)
economics.wacc	0.07 real	indicative — unverified (judgement)
economics.contingency	0.1 fraction of capex	indicative — unverified (judgement)
economics.capexLow	-0.3 fraction (AACE Class 5)	indicative — unverified (source)
economics.capexHigh	0.5 fraction (AACE Class 5)	indicative — unverified (source)
economics.carbonPrice	by country	indicative — unverified (source). The country default comes from the scheme records country.carbonPrice.euEts and country.carbonPrice.ukEts (sourced); Switzerland is unresearched.
economics.carbonEscalation	0.02 real per yr	indicative — unverified (source)
economics.capexFactor	1 × every capex line	indicative — unverified (judgement). Tornado driver "capex"; range the AACE Class 5 band (-30 % / +50 %).
economics.bessCapexFactor	1 × battery capex	indicative — unverified (judgement). Tornado driver "battery capex". No source.
economics.gridCapexFactor	1 × grid connection capex	indicative — unverified (judgement). Tornado driver "grid connection capex". Scales the grid connection charge and the grid cable route only, not the substation, which moves with the capex driver (owner, 2026-10-26). No source.
economics.elecEscalation	0 real per yr	indicative — unverified (source)
economics.gasEscalation	0 real per yr	indicative — unverified (source)
economics.salvageFraction	0.5 of remaining book value when removed	indicative — unverified (judgement)
economics.revenueExport	no	indicative — unverified (judgement)
economics.displayPeriods	10; 15	indicative — unverified (judgement). Walkthrough 2026-10-21: several evaluation periods at once (for example 10, 15 and 20 years). The scoring period (economics.periodYears) sets the cost axis.

Default used	Value	Basis and note
weights.cost	35 raw 0–100	indicative — unverified (judgement)
weights.carbon	25 raw 0–100	indicative — unverified (judgement)
weights.ttp	25 raw 0–100	indicative — unverified (judgement)
weights.deliverability	15 raw 0–100	indicative — unverified (judgement)
deliverability.emergingPenalty	15 points per 100 % share	indicative — unverified (judgement)
deliverability.nascentPenalty	40 points per 100 % share	indicative — unverified (judgement)
deliverability.majorFlagPenalty	8 points per major permitting flag	indicative — unverified (judgement)
deliverability.minorFlagPenalty	3 points per minor permitting flag	indicative — unverified (judgement)
deliverability.gasUnknownPenalty	10 points	indicative — unverified (judgement)
deliverability.hydrogenPenalty	10 points	indicative — unverified (judgement)
deliverability.leadTimePenalty	5 points per technology	indicative — unverified (judgement)
deliverability.leadTimeThresholdM	24 months procurement	indicative — unverified (judgement)
deliverability.pooledBackupRisk	minor	indicative — unverified (judgement). Client-perceived risk of backup pooled across the campus (OPEN-QUESTIONS 48, 2026-10-22): by default the same penalty as a minor permitting flag; the engineer can raise it to major for a project.
settings.generationCap	2000 options	indicative — unverified (run setting)
settings.sampleAboveCap	yes	indicative — unverified (judgement)
settings.gridHandling	unmet	indicative — unverified (judgement). Show unmet load
settings.runHourHandling	flag	indicative — unverified (judgement). Flag only
settings.costMetric	lcoe	indicative — unverified (judgement)
settings.carbonMetric	location	indicative — unverified (judgement)
settings.absoluteScaling	no	indicative — unverified (judgement). Hook only in this slice.
settings.rankCompliantFirst	no	indicative — unverified (judgement). On: options whose resilience status is "Does not meet" rank after the others; scores do not change.
settings.hideDominated	no	indicative — unverified (judgement)
settings.windSiteThresholdHa	20 ha	indicative — unverified (judgement)
settings.bridgeBessHours	24 h	indicative — unverified (judgement)
settings.timelineMonths	120 months	indicative — unverified (run setting). Time-to-power horizon.
settings.rentalLeadMonths	1 months	indicative — unverified (judgement). Rented bridging plant is hired this many months before it is first needed (installation and commissioning), not from the month it could first arrive (known issue 15, work package 11). No source.
settings.maxDispatchYears	15 years	indicative — unverified (run setting). Upper limit on the distinct 8760-h dispatch years; each option is dispatched until nothing changes (full load after the last phase and its ramp, grid steps, plant changes) and the last year then repeats. Was 8 until work package 11 (known issue 17): a long programme repeated a part-built year. A run flag says when the limit cuts an option short.
settings.screeningModel	full	indicative — unverified (run setting). OPEN-QUESTIONS 23: full-year hourly for every option by default; representative days (the design day, the dearest day and three clustered typical days per month, about 50 days) as an optional faster screening.
settings.repDaysPerMonth	3 typical days a month (plus the dearest day and the design day)	indicative — unverified (run setting). Screening model: the days of each month are grouped into this many clusters on price, solar, wind, grid carbon and temperature; each cluster is represented by its most central day, weighted by its size.
settings.verifyTop	10 options verified with the full year	indicative — unverified (run setting). OPEN-QUESTIONS 23: the best-ranked options re-run hour by hour when the screening model is on.
settings.autonomyBasis	critical load	indicative — unverified (judgement)
settings.layoutCellM	2 m raster cell	indicative — unverified (run setting). Auto-layout raster cell size; chosen for the §17 layout timing, not from a source.
settings.layoutCellLargeM	5 m raster cell (large sites)	indicative — unverified (run setting)
settings.siteLayoutScope	auto	indicative — unverified (run setting). all: every option, always.
settings.siteLayoutWorkLimit	20000000 block2 (ground blocks squared, summed over the options)	indicative — unverified (judgement). T1 (14 ha, 25 MW buildings, about 21 blocks an option): about 1 million, laid out in 5 s. Pego (150 ha, nine 100 MW buildings, about 230 blocks an option): about 105 million, 9 minutes on one core (2026-10-24).

Default used	Value	Basis and note
settings.layoutLargeSiteHa	50 ha gross site area above which the large cell is used	indicative — unverified (run setting)

3. Results

2,000 options were modelled hour by hour for a full year (8,760 hours) and ranked by a weighted score: cost 35 %, carbon 25 %, time-to-power 25 %, deliverability 15 % (cost measured as LCOE, carbon location-based). Options that meet the power need in every hour (power adequacy) rank first: 1,610 of 2,000 are adequate.

Leader: #1 Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · retained-plant backup · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 %, score 89.26, LCOE €159/MWh, 175 gCO2e/kWh (location-based), full load 2028-10.

Top five strategies

The best-ranked option of each distinct strategy (grid approach, bridging, permanent on-site generation and backup), skipping options with the same results as a better-ranked one. Pros and cons are judged against the 2,000 options in this run.

#1 Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · retained-plant backup · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 %

Score	LCOE €/MWh	Capex	Carbon g/kWh	First power	Full load	Resilience	Adequacy
89.26	159	€1.17 bn	175	2029-01	2028-10	Meets with flags	Adequate

- Pro: Deliverability in the best 1 % (89 points)
- Pro: LCOE in the best 8 % (€159/MWh)

Main flags: Aggregated combustion thermal input 2415 MWth >= 50 MWth (IED threshold)

#2 Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · Diesel N+1 · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 %

Score	LCOE €/MWh	Capex	Carbon g/kWh	First power	Full load	Resilience	Adequacy
89.17	160	€1.2 bn	175	2029-01	2029-04	Meets with flags	Adequate

- Pro: Deliverability in the best 1 % (89 points)
- Pro: LCOE in the best 9 % (€160/MWh)

Main flags: Aggregated combustion thermal input 2750 MWth >= 50 MWth (IED threshold)

#4 Grid + gas recips bridge -> keep shortfall share as prime, rest as backup · retained-plant backup · Evaporative-adiabatic · BESS for the shortfall · PPA hybrid 100 %

Score	LCOE €/MWh	Capex	Carbon g/kWh	First power	Full load	Resilience	Adequacy
89.12	159	€1.14 bn	179	2029-01	2028-10	Meets with flags	Adequate

- Pro: Deliverability in the best 1 % (89 points)
- Pro: LCOE in the best 8 % (€159/MWh)

Main flags: Aggregated combustion thermal input 2318 MWth >= 50 MWth (IED threshold)

#6 Grid + gas recips bridge -> retain as backup · Fuel cells top-up · retained-plant backup · Evaporative-adiabatic · PV 25 MWp · BESS for the shortfall · PPA hybrid 100 %

Score	LCOE €/MWh	Capex	Carbon g/kWh	First power	Full load	Resilience	Adequacy
88.78	161	€1.28 bn	176	2029-01	2028-10	Meets with flags	Adequate

- Pro: Deliverability in the best 8 % (89 points)
- Pro: LCOE in the best 10 % (€161/MWh)

Main flags: Aggregated combustion thermal input 2318 MWth >= 50 MWth (IED threshold)

#9 Grid + gas recips bridge -> retain as backup · Fuel cells top-up · HVO N+1 · Evaporative-adiabatic · PV 25 MWp · BESS 2 h · PPA hybrid 100 %

Score	LCOE €/MWh	Capex	Carbon g/kWh	First power	Full load	Resilience	Adequacy
88.54	163	€1.37 bn	176	2029-01	2028-10	Meets with flags	Adequate

- Pro: Deliverability in the best 7 % (89 points)

Main flags: Aggregated combustion thermal input 2653 MWth >= 50 MWth (IED threshold)

Top five by evaluation period

LCOE and NPV of net cost over each period; the 20-year period sets the ranking.

Option	LCOE 10 yr	NPV 10 yr	LCOE 15 yr	NPV 15 yr	LCOE 20 yr	NPV 20 yr
#1 Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · retained-plant backup · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 %	€173/MWh	€3.66 bn	€163/MWh	€5.51 bn	€159/MWh	€6.79 bn
#2 Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · Diesel N+1 · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 %	€174/MWh	€3.68 bn	€164/MWh	€5.53 bn	€160/MWh	€6.82 bn
#4 Grid + gas recips bridge -> keep shortfall share as prime, rest as backup · retained-plant backup · Evaporative-adiabatic · BESS for the shortfall · PPA hybrid 100 %	€173/MWh	€3.66 bn	€163/MWh	€5.5 bn	€159/MWh	€6.79 bn
#6 Grid + gas recips bridge -> retain as backup · Fuel cells top-up · retained-plant backup · Evaporative-adiabatic · PV 25 MWp · BESS for the shortfall · PPA hybrid 100 %	€177/MWh	€3.73 bn	€166/MWh	€5.6 bn	€161/MWh	€6.9 bn
#9 Grid + gas recips bridge -> retain as backup · Fuel cells top-up · HVO N+1 · Evaporative-adiabatic · PV 25 MWp · BESS 2 h · PPA hybrid 100 %	€179/MWh	€3.79 bn	€168/MWh	€5.67 bn	€163/MWh	€6.98 bn

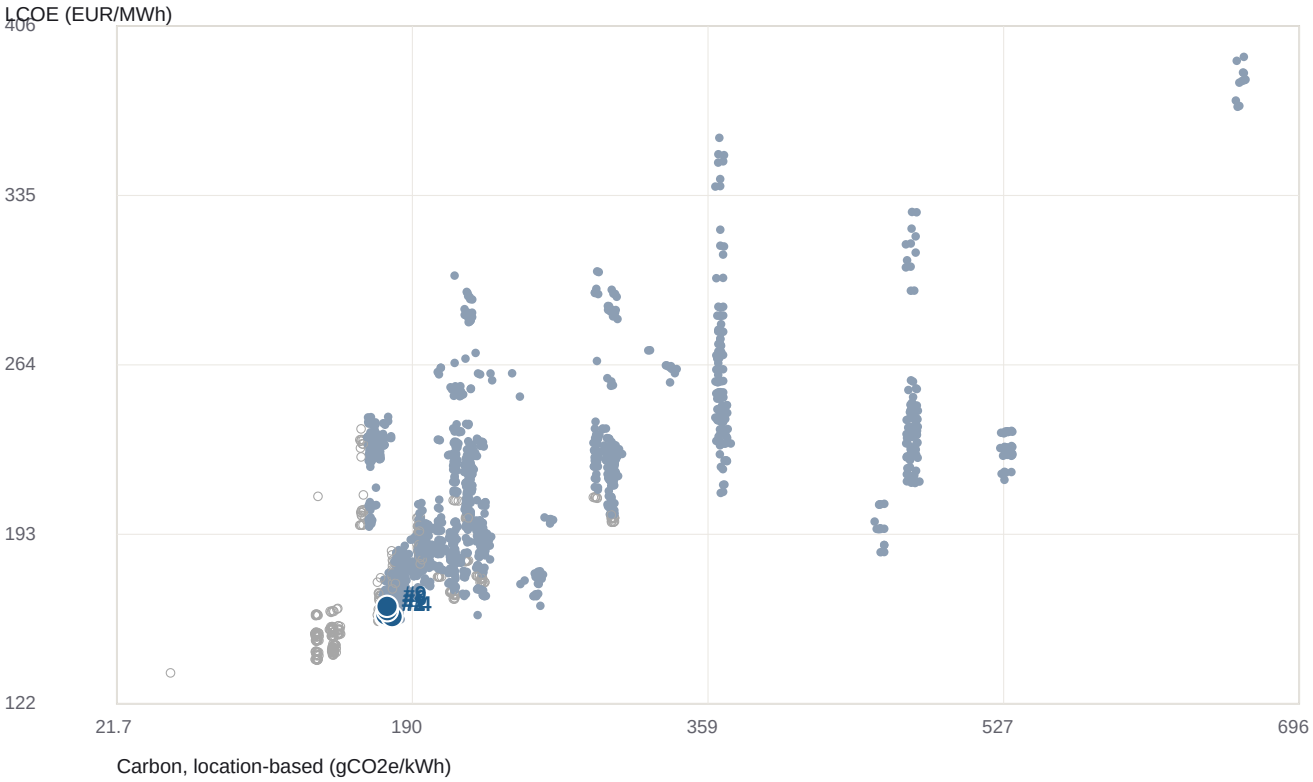
Ranked list (top 20)

#	Option	Score	LCOE €/MWh	Capex	g/kWh	Full load	Adequate
1	Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · retained-plant backup · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 %	89.26	159	€1.17 bn	175	2028-10	yes
2	Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · Diesel N+1 · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 %	89.17	160	€1.2 bn	175	2029-04	yes
3	Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · Diesel N+1 · Evaporative-adiabatic · PPA hybrid 100 %	89.14	159	€1.18 bn	176	2029-04	yes
4	Grid + gas recips bridge -> keep shortfall share as prime, rest as backup · retained-plant backup · Evaporative-adiabatic · BESS for the shortfall · PPA hybrid 100 %	89.12	159	€1.14 bn	179	2028-10	yes
5	Grid + gas recips bridge -> keep shortfall share as prime, rest as backup · Gas CHP-trigen top-up · retained-plant backup · Evaporative-adiabatic · PV 42 MWp · BESS for the shortfall · PPA hybrid 100 %	88.83	160	€1.24 bn	181	2028-10	yes
6	Grid + gas recips bridge -> retain as backup · Fuel cells top-up · retained-plant backup · Evaporative-adiabatic · PV 25 MWp · BESS for the shortfall · PPA hybrid 100 %	88.78	161	€1.28 bn	176	2028-10	yes
7	Grid + gas recips bridge -> retain as backup · Fuel cells top-up · retained-plant backup · Evaporative-adiabatic · PV 42 MWp · BESS 2 h · PPA hybrid 100 %	88.66	162	€1.35 bn	175	2028-10	yes

#	Option	Score	LCOE €/MWh	Capex	g/kWh	Full load	Adequate
8	Grid + gas recips bridge -> keep shortfall share as prime, rest as backup · Fuel cells top-up · retained-plant backup · Evaporative-adiabatic · PV 42 MWp · BESS for the shortfall · PPA hybrid 100 %	88.56	161	€1.3 bn	181	2028-10	yes
9	Grid + gas recips bridge -> retain as backup · Fuel cells top-up · HVO N+1 · Evaporative-adiabatic · PV 25 MWp · BESS 2 h · PPA hybrid 100 %	88.54	163	€1.37 bn	176	2028-10	yes
10	Grid + gas recips bridge -> keep shortfall share as prime, rest as backup · Fuel cells top-up · Diesel N+1 · Evaporative-adiabatic · BESS for the shortfall · PPA hybrid 100 %	88.42	162	€1.29 bn	183	2028-10	yes
11	Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · HVO N+1 pooled · Evaporative-adiabatic · PV 42 MWp · PPA hybrid 100 %	88.35	159	€1.16 bn	174	2029-04	yes
12	Grid + gas recips bridge -> keep shortfall share as prime, rest as backup · Fuel cells top-up · HVO N+1 · Evaporative-adiabatic · PV 25 MWp · BESS 2 h · PPA hybrid 100 %	88.3	163	€1.37 bn	182	2028-10	yes
13	Grid + gas recips bridge -> keep shortfall share as prime, rest as backup · HVO N+1 · Evaporative-adiabatic · BESS 2 h · PPA hybrid 100 %	88.2	161	€1.23 bn	178	2033-01	yes
14	Grid + gas recips bridge -> keep shortfall share as prime, rest as backup · Gas CHP-trigen top-up · retained-plant backup · Evaporative-adiabatic · PV 33 MWp	88.11	165	€1.18 bn	181	2028-10	yes
15	Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · HVO N+1 · Evaporative-adiabatic · PV 42 MWp · BESS 4 h · PPA hybrid 100 %	88.01	169	€1.57 bn	174	2029-04	yes
16	Grid + gas recips bridge -> keep shortfall share as prime, rest as backup · Gas CHP-trigen top-up · HVO N+1 pooled · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 %	88.01	159	€1.17 bn	181	2029-04	yes
17	Grid + rented temporary generation bridge -> keep shortfall share as prime, rest as backup · retained-plant backup · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 %	88.01	167	€99.3 m	180	2028-12	yes
18	Grid + gas recips bridge -> retain as backup · Fuel cells top-up · retained-plant backup · Evaporative-adiabatic · PV 25 MWp	87.98	166	€1.23 bn	178	2028-10	yes
19	Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · HVO N+1 · Evaporative-adiabatic · PV 25 MWp · BESS 2 h	87.92	169	€1.31 bn	175	2029-04	yes
20	Grid + gas recips bridge -> retain as backup · Fuel cells top-up · Diesel N+1 · Evaporative-adiabatic · PV 42 MWp	87.91	167	€1.28 bn	178	2028-10	yes

4. Trade-offs

Cost (LCOE, up) against carbon (location-based, across) for every option. Filled dots are adequate options, hollow dots are not; the top five are numbered by rank.



5. Compare

No options were chosen for comparison when the report was made (tick "Compare" on a card or use "Add to compare").

6. Dispatch: energy balance

Where each top-five option gets its energy in the last modelled year (steady state, every phase at full load) and in the first year, in GWh. Plant is on-site generation (bridging, permanent and backup on primary hours); flexibility is load turned down; unmet is demand not served. Hour-by-hour charts are in the Dispatch tab.

Last modelled year, steady state (model year 8, from 2034-01)

Option	Demand	Grid	PPA	On-site RE	Plant	Battery	Flex	Unmet
#1 Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · retained-plant backup · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 %	6,070	2,050	3,930	23.8	72.4	0	0	0
#2 Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · Diesel N+1 · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 %	6,070	2,050	3,930	23.8	72.4	0	0	0
#4 Grid + gas recips bridge -> keep shortfall share as prime, rest as backup · retained-plant backup · Evaporative-adiabatic · BESS for the shortfall · PPA hybrid 100 %	6,070	1,980	3,950	0	146	0	0	0
#6 Grid + gas recips bridge -> retain as backup · Fuel cells top-up · retained-plant backup · Evaporative-adiabatic · PV 25 MWp · BESS for the shortfall · PPA hybrid 100 %	6,070	1,930	3,930	23.8	188	0	0	0

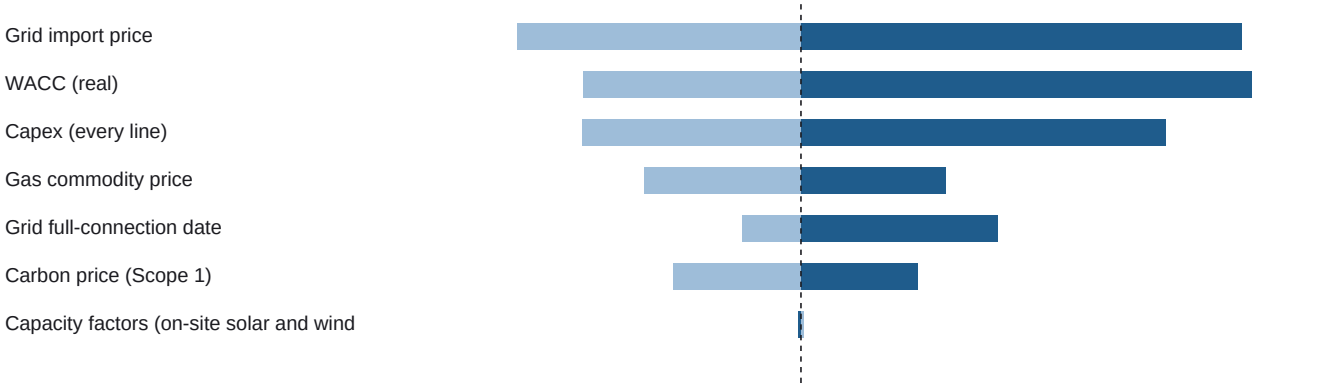
Option	Demand	Grid	PPA	On-site RE	Plant	Battery	Flex	Unmet
#9 Grid + gas recips bridge -> retain as backup · Fuel cells top-up · HVO N+1 · Evaporative-adiabatic · PV 25 MWp · BESS 2 h · PPA hybrid 100 %	6,070	1,940	3,930	23.8	186	71.1	0	0

First year with load (each option, see the first column)

Option	Demand	Grid	PPA	On-site RE	Plant	Battery	Flex	Unmet
#1 Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · retained-plant backup · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 % (model year 3, from 2029-01)	675	0	0	23.8	652	0	0	0
#2 Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · Diesel N+1 · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 % (model year 3, from 2029-01)	675	0	0	23.8	652	0	0	0
#4 Grid + gas recips bridge -> keep shortfall share as prime, rest as backup · retained-plant backup · Evaporative-adiabatic · BESS for the shortfall · PPA hybrid 100 % (model year 3, from 2029-01)	679	0	0	0	679	0	0	0
#6 Grid + gas recips bridge -> retain as backup · Fuel cells top-up · retained-plant backup · Evaporative-adiabatic · PV 25 MWp · BESS for the shortfall · PPA hybrid 100 % (model year 3, from 2029-01)	679	0	0	23.8	656	0	0	0
#9 Grid + gas recips bridge -> retain as backup · Fuel cells top-up · HVO N+1 · Evaporative-adiabatic · PV 25 MWp · BESS 2 h · PPA hybrid 100 % (model year 3, from 2029-01)	679	0	0	23.8	655	0.464	0	0

7. Sensitivity

#1 Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · retained-plant backup · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 %: LCOE at each driver's low and high value, everything else held; base €158.9/MWh. 7 drivers tested; largest swing first. Ranges are registry records, indicative — unverified.



LCOE: light bar = low value, dark bar = high value; dashed line = base. Axis €147.5/MWh to €175.8/MWh.

Driver	Low	LCOE	High	LCOE	Score low · high
Grid import price	× 0.7	€149.4/MWh	× 1.5	€173.5/MWh	90.68 · 87.03
WACC (real)	4 %	€151.6/MWh	12 %	€173.9/MWh	90.25 · 87.23
Capex (every line)	× 0.7	€151.6/MWh	× 1.5	€171/MWh	90.25 · 87.62
Gas commodity price	× 0.7	€153.7/MWh	× 1.3	€163.7/MWh	89.74 · 88.74
Grid full-connection date	-6 months	€156.9/MWh	+24 months	€165.4/MWh	89.8 · 87.04
Carbon price (Scope 1)	× 0.5	€154.6/MWh	× 1.5	€162.7/MWh	89.65 · 88.86
Capacity factors (on-site solar and wind)	× 0.85	€159/MWh	× 1.15	€158.8/MWh	89.25 · 89.28

Breakevens

The value of each driver, nearest its current value, at which the leader (#1) and a competitor swap places ("no flip" within the driver's range otherwise). Competitors: #2, then the best option of each other strategy; drivers: the 8 that move the leader's weighted score most. Each side of the current value is checked at its middle and its end and the first change of order narrowed to 2 % of the range (whole-number drivers: to 1); a swap that reverses between two checked points is not seen.

Driver (range)	#2	#4	#6	#9	#10
Electricity price escalation (-2.0 %/yr to +2.0 %/yr)	lower: no flip; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip
Grid import price (× 0.7 to × 1.5)	lower: no flip; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip
Evaluation period (15 years to 25 years)	lower: no flip; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip
WACC (real) (4 % to 12 %)	lower: no flip; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip
Grid full-connection date (-6 months to +24 months)	earlier: no flip; later: +9 months (2033-10)	earlier: no flip; later: no flip	earlier: no flip; later: no flip	earlier: no flip; later: no flip	earlier: no flip; later: no flip
Capex (every line) (× 0.7 to × 1.5)	lower: no flip; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip
Grid carbon intensity (× 0.8 to × 1.2)	lower: no flip; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip
IT load (all phases) (× 0.8 to × 1.2)	lower: no flip; higher: no flip	lower: × 0.978; higher: no flip	lower: × 0.966; higher: no flip	lower: no flip; higher: no flip	lower: no flip; higher: no flip

- #2 Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · Diesel N+1 · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 %
- #4 Grid + gas recips bridge -> keep shortfall share as prime, rest as backup · retained-plant backup · Evaporative-adiabatic · BESS for the shortfall · PPA hybrid 100 %
- #6 Grid + gas recips bridge -> retain as backup · Fuel cells top-up · retained-plant backup · Evaporative-adiabatic · PV 25 MWp · BESS for the shortfall · PPA hybrid 100 %
- #9 Grid + gas recips bridge -> retain as backup · Fuel cells top-up · HVO N+1 · Evaporative-adiabatic · PV 25 MWp · BESS 2 h · PPA hybrid 100 %
- #10 Grid + gas recips bridge -> keep shortfall share as prime, rest as backup · Fuel cells top-up · Diesel N+1 · Evaporative-adiabatic · BESS for the shortfall · PPA hybrid 100 %

Nearest flips: #4 would lead if IT load fell to × 0.978; #6 would lead if IT load fell to × 0.966; #2 would lead if the grid full-connection date slipped by 9 months or more.

Uncertainty (Monte Carlo)

300 Latin hypercube samples (seed 20261025) of 13 inputs drawn together, each triangular between its low and high value around the current one; grid and gas price correlated, and the capex factors. The 20 leading options re-run in each sample with representative-day dispatch and put in the ranking's order. #1 leads in 49 % of samples and is in the top three in 94 %; the robust pick (most often within 1 % of the best score) is #1, the leader itself. Representative-day dispatch makes adequacy and the order of the options approximate: now and then a full-year run would put a different option first in a sample.

Option	Leads · top 3	Within 1 %	LCOE P10 · P50 · P90	Carbon P50	Full load P10 – P90	Adequate
#1	49 % · 94 %	99.0 %	€151/MWh · €167/MWh · €182/MWh	185 g/kWh	2028-10 – 2028-10	100 %
#2	30 % · 69 %	92.0 %	€152/MWh · €167/MWh · €182/MWh	185 g/kWh	2028-10 – 2032-09	100 %
#3	1 % · 38 %	92.0 %	€152/MWh · €167/MWh · €182/MWh	186 g/kWh	2028-10 – 2032-09	100 %
#4	5 % · 59 %	98.7 %	€151/MWh · €167/MWh · €182/MWh	188 g/kWh	2028-10 – 2028-10	100 %

Option	Leads · top 3	Within 1 %	LCOE P10 · P50 · P90	Carbon P50	Full load P10 – P90	Adequate
#5	0 % · 6 %	88.0 %	€153/MWh · €169/MWh · €183/MWh	191 g/kWh	2028-10 – 2028-10	100 %
#6	1 % · 8 %	84.7 %	€153/MWh · €169/MWh · €185/MWh	185 g/kWh	2028-10 – 2028-10	100 %
#7	0 % · 1 %	80.3 %	€154/MWh · €171/MWh · €187/MWh	184 g/kWh	2028-10 – 2028-10	100 %
#8	0 % · 0 %	66.7 %	€153/MWh · €169/MWh · €185/MWh	190 g/kWh	2028-10 – 2028-10	100 %
#9	0 % · 0 %	64.0 %	€155/MWh · €171/MWh · €187/MWh	184 g/kWh	2028-10 – 2032-10	100 %
#10	0 % · 0 %	42.3 %	€154/MWh · €170/MWh · €186/MWh	192 g/kWh	2028-10 – 2032-10	100 %
#11	0 % · 0 %	6.3 %	€151/MWh · €167/MWh · €182/MWh	184 g/kWh	2028-10 – 2032-09	100 %
#12	0 % · 0 %	28.3 %	€156/MWh · €172/MWh · €187/MWh	190 g/kWh	2028-10 – 2032-10	100 %
#13	0 % · 0 %	74.0 %	€153/MWh · €169/MWh · €184/MWh	187 g/kWh	2028-10 – 2032-12	100 %
#14	5 % · 8 %	29.3 %	€153/MWh · €174/MWh · €200/MWh	193 g/kWh	2028-10 – 2028-10	100 %
#15	0 % · 0 %	8.0 %	€160/MWh · €177/MWh · €195/MWh	183 g/kWh	2028-10 – 2032-09	100 %
#16	0 % · 0 %	0.0 %	€151/MWh · €167/MWh · €181/MWh	191 g/kWh	2028-10 – 2032-09	100 %
#17	8 % · 10 %	32.3 %	€161/MWh · €173/MWh · €187/MWh	192 g/kWh	2028-12 – 2028-12	100 %
#18	1 % · 4 %	24.3 %	€153/MWh · €176/MWh · €202/MWh	187 g/kWh	2028-10 – 2028-10	100 %
#19	0 % · 2 %	21.3 %	€156/MWh · €178/MWh · €205/MWh	185 g/kWh	2028-10 – 2032-09	100 %
#20	0 % · 2 %	21.3 %	€155/MWh · €176/MWh · €202/MWh	187 g/kWh	2028-10 – 2032-10	100 %

8. Flags

Every flag raised on the top-five options. Flags never hide, demote or resize an option (the one exception agreed by the owner: power-adequate options rank first). Site-fit flags (Site tab, Beta) are not listed.

#1 Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · retained-plant backup · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 %

Severity	Category	Flag
major	Permitting	Aggregated combustion thermal input 2415 MWth >= 50 MWth (IED threshold). Screening flag, not a permitting determination.
minor	Resilience	Backup depends on gas network availability during a grid outage: plant that does not burn gas covers 0.0 of 916.3 MW critical load
minor	Grid	PPA limited by import capacity: 6,775,345 MWh (16.7 % of the 40,580,322 MWh undelivered) of contracted PPA volume over the 20-year evaluation period could not be delivered because the grid connection was full. Its settlement, a net cost of €112 m (present value €27.8 m), is reported as "PPA hedge (not counted)": it is not in the cost ledger, LCOE, NPV or score, because PPA settlement counts only on energy delivered over the grid connection.
minor	Permitting	Individual combustion units of 10.7 MWth (Gas CHP / trigeneration), 22.7 MWth (Gas reciprocating engines) are within 1–50 MWth (medium combustion plant rules). Screening flag, not a permitting determination.
minor	Weather	Weather: uploaded EPW file "NLD_Amsterdam062400_IWEC.epw" (AMSTERDAM, 1982–1999 (months from several years)), 189 km from the pin; sized to n = 20: 34.6 °C dry bulb and 24.5 °C wet bulb; check at n = 50: 36 °C and 24.5 °C, from the file's ASHRAE extremes (Climate Design Data 2009 ASHRAE Handbook): n = 20 and n = 50-year maximum dry bulb, and the extreme maximum wet bulb (no return period given for wet bulb). Warning: more than 50 km from the pin: check that its climate is representative
info	Grid	PPA volume above the load: 33,804,977 MWh (83.3 % of the 40,580,322 MWh undelivered) of contracted PPA volume over the 20-year evaluation period exceeded the load still to serve in its hour. Its settlement, a net cost of €112 m (present value €27.8 m), is reported as "PPA hedge (not counted)": it is not in the cost ledger, LCOE, NPV or score, because PPA settlement counts only on energy delivered over the grid connection.
info	Grid	Grid connection charge assumed €0 (the default, indicative — unverified); at the standard rate of €120k per MW (EHV) the 950 MW contracted would cost about €114.0 m
info	Permitting	Permitting outlook: Gas CHP / trigeneration (prime; IED): uncertain; ready month 27, needed month 78, 51 months' float; Gas reciprocating engines (bridging; IED): uncertain; ready month 21, needed month 24, 3 months' float. Screening flag, not a permitting determination.
info	Fuel and utilities	Gas capacity and date assumed, not confirmed
info	Data	Result depends on C-tagged (indicative — unverified) parameters and synthetic profiles
info	Assumed default	Assumed Analysis start: 2027-01, default (judgement, indicative — unverified)
info	Assumed default	Assumed Average IT utilisation: 0.8 (fraction of installed IT), default (judgement, indicative — unverified)

Severity	Category	Flag
info	Assumed default	Assumed Ramp: share at energisation: 0.25 (fraction of phase IT at energisation), default (judgement, indicative — unverified)
info	Assumed default	Assumed Load shape: daily, default (judgement, indicative — unverified)
info	Assumed default	Assumed Grid non-commodity adder: Netherlands default (source, indicative — unverified)
info	Assumed default	Assumed Electricity price escalation: 0 (real per yr), default (source, indicative — unverified)
info	Assumed default	Assumed Gas commodity price: Netherlands default (source, indicative — unverified)
info	Assumed default	Assumed Carbon price (Scope 1): Netherlands default (source, indicative — unverified)
info	Assumed default	Assumed WACC (real): 0.07 real, default (judgement, indicative — unverified)
info	Assumed default	Assumed Evaluation period: 20 yr, default (judgement, indicative — unverified)
info	Assumed default	Assumed UPS ride-through: 5 (min at full IT), default (judgement, indicative — unverified)
info	Assumed default	Assumed PPA type: hybrid, default (judgement, indicative — unverified)
info	Design check	Design check at n = 50 (24.5 °C): peak PUE 1.292, design facility load 983.8 MW (+0.0 MW over the n = 20 sizing); not used for sizing
info	Assumed default	Technology, cost, carbon and other defaults: 550 of 574 registry records are indicative — unverified (PDF report section 2; user guide appendix)

#2 Grid + gas recips bridge -> retain as backup · Gas CHP-trigen top-up · Diesel N+1 · Evaporative-adiabatic · PV 25 MWp · PPA hybrid 100 %

Severity	Category	Flag
major	Permitting	Aggregated combustion thermal input 2750 MWth >= 50 MWth (IED threshold). Screening flag, not a permitting determination.
minor	Resilience	Backup depends on gas network availability during a grid outage: plant that does not burn gas covers 154.0 of 916.3 MW critical load
minor	Grid	PPA limited by import capacity: 6,775,345 MWh (16.7 % of the 40,580,322 MWh undelivered) of contracted PPA volume over the 20-year evaluation period could not be delivered because the grid connection was full. Its settlement, a net cost of €112 m (present value €27.8 m), is reported as "PPA hedge (not counted)": it is not in the cost ledger, LCOE, NPV or score, because PPA settlement counts only on energy delivered over the grid connection.
minor	Permitting	Individual combustion units of 10.7 MWth (Gas CHP / trigeneration), 22.7 MWth (Gas reciprocating engines), 7.2 MWth (Diesel standby gensets) are within 1–50 MWth (medium combustion plant rules). Screening flag, not a permitting determination.
minor	Permitting	Permitting outlook: Gas CHP / trigeneration (prime; IED): uncertain; ready month 27, needed month 78, 51 months' float; Gas reciprocating engines (bridging; IED): uncertain; ready month 21, needed month 24, 3 months' float; Diesel standby gensets (backup; IED): likely; ready month 27, needed month 24, 3 months late. Screening flag, not a permitting determination.
minor	Weather	Weather: uploaded EPW file "NLD_Amsterdam062400_IWEC.epw" (AMSTERDAM, 1982–1999 (months from several years)), 189 km from the pin; sized to n = 20: 34.6 °C dry bulb and 24.5 °C wet bulb; check at n = 50: 36 °C and 24.5 °C, from the file's ASHRAE extremes (Climate Design Data 2009 ASHRAE Handbook): n = 20 and n = 50-year maximum dry bulb, and the extreme maximum wet bulb (no return period given for wet bulb). Warning: more than 50 km from the pin: check that its climate is representative
info	Grid	PPA volume above the load: 33,804,977 MWh (83.3 % of the 40,580,322 MWh undelivered) of contracted PPA volume over the 20-year evaluation period exceeded the load still to serve in its hour. Its settlement, a net cost of €112 m (present value €27.8 m), is reported as "PPA hedge (not counted)": it is not in the cost ledger, LCOE, NPV or score, because PPA settlement counts only on energy delivered over the grid connection.
info	Grid	Grid connection charge assumed €0 (the default, indicative — unverified); at the standard rate of €120k per MW (EHV) the 950 MW contracted would cost about €114.0 m
info	Fuel and utilities	Gas capacity and date assumed, not confirmed
info	Data	Result depends on C-tagged (indicative — unverified) parameters and synthetic profiles
info	Assumed default	Assumed Analysis start: 2027-01, default (judgement, indicative — unverified)
info	Assumed default	Assumed Average IT utilisation: 0.8 (fraction of installed IT), default (judgement, indicative — unverified)
info	Assumed default	Assumed Ramp: share at energisation: 0.25 (fraction of phase IT at energisation), default (judgement, indicative — unverified)
info	Assumed default	Assumed Load shape: daily, default (judgement, indicative — unverified)
info	Assumed default	Assumed Grid non-commodity adder: Netherlands default (source, indicative — unverified)
info	Assumed default	Assumed Electricity price escalation: 0 (real per yr), default (source, indicative — unverified)
info	Assumed default	Assumed Gas commodity price: Netherlands default (source, indicative — unverified)
info	Assumed default	Assumed Carbon price (Scope 1): Netherlands default (source, indicative — unverified)

Severity	Category	Flag
info	Assumed default	Assumed WACC (real): 0.07 real, default (judgement, indicative — unverified)
info	Assumed default	Assumed Evaluation period: 20 yr, default (judgement, indicative — unverified)
info	Assumed default	Assumed Backup autonomy: 48 h, default (judgement, indicative — unverified)
info	Assumed default	Assumed Generator test hours: 6 (h/yr per unit), default (judgement, indicative — unverified)
info	Assumed default	Assumed UPS ride-through: 5 (min at full IT), default (judgement, indicative — unverified)
info	Assumed default	Assumed PPA type: hybrid, default (judgement, indicative — unverified)
info	Design check	Design check at n = 50 (24.5 °C): peak PUE 1.292, design facility load 983.8 MW (+0.0 MW over the n = 20 sizing); not used for sizing
info	Assumed default	Technology, cost, carbon and other defaults: 550 of 574 registry records are indicative — unverified (PDF report section 2; user guide appendix)

#4 Grid + gas recips bridge -> keep shortfall share as prime, rest as backup · retained-plant backup · Evaporative-adiabatic · BESS for the shortfall · PPA hybrid 100 %

Severity	Category	Flag
major	Permitting	Aggregated combustion thermal input 2318 MWth >= 50 MWth (IED threshold). Screening flag, not a permitting determination.
minor	Resilience	Backup depends on gas network availability during a grid outage: plant that does not burn gas covers 0.0 of 916.3 MW critical load
minor	Grid	PPA limited by import capacity: 6,858,496 MWh (17.0 % of the 40,292,063 MWh undelivered) of contracted PPA volume over the 20-year evaluation period could not be delivered because the grid connection was full. Its settlement, a net cost of €114 m (present value €28.7 m), is reported as "PPA hedge (not counted)": it is not in the cost ledger, LCOE, NPV or score, because PPA settlement counts only on energy delivered over the grid connection.
minor	Grid	Battery for the shortfall cannot recharge from the grid headroom on the design day from month 80: it does not count as firm supply against the design load
minor	Permitting	Individual combustion units of 22.7 MWth (Gas reciprocating engines) are within 1–50 MWth (medium combustion plant rules). Screening flag, not a permitting determination.
minor	Weather	Weather: uploaded EPW file "NLD_Amsterdam062400_IWEC.epw" (AMSTERDAM, 1982–1999 (months from several years)), 189 km from the pin; sized to n = 20: 34.6 °C dry bulb and 24.5 °C wet bulb; check at n = 50: 36 °C and 24.5 °C, from the file's ASHRAE extremes (Climate Design Data 2009 ASHRAE Handbook): n = 20 and n = 50-year maximum dry bulb, and the extreme maximum wet bulb (no return period given for wet bulb). Warning: more than 50 km from the pin: check that its climate is representative
info	Grid	PPA volume above the load: 33,433,567 MWh (83.0 % of the 40,292,063 MWh undelivered) of contracted PPA volume over the 20-year evaluation period exceeded the load still to serve in its hour. Its settlement, a net cost of €114 m (present value €28.7 m), is reported as "PPA hedge (not counted)": it is not in the cost ledger, LCOE, NPV or score, because PPA settlement counts only on energy delivered over the grid connection.
info	Grid	Grid connection charge assumed €0 (the default, indicative — unverified); at the standard rate of €120k per MW (EHV) the 950 MW contracted would cost about €114.0 m
info	Permitting	Permitting outlook: Gas reciprocating engines (bridging; IED): uncertain; ready month 21, needed month 24, 3 months' float. Screening flag, not a permitting determination.
info	Fuel and utilities	Gas capacity and date assumed, not confirmed
info	Data	Result depends on C-tagged (indicative — unverified) parameters and synthetic profiles
info	Assumed default	Assumed Analysis start: 2027-01, default (judgement, indicative — unverified)
info	Assumed default	Assumed Average IT utilisation: 0.8 (fraction of installed IT), default (judgement, indicative — unverified)
info	Assumed default	Assumed Ramp: share at energisation: 0.25 (fraction of phase IT at energisation), default (judgement, indicative — unverified)
info	Assumed default	Assumed Load shape: daily, default (judgement, indicative — unverified)
info	Assumed default	Assumed Grid non-commodity adder: Netherlands default (source, indicative — unverified)
info	Assumed default	Assumed Electricity price escalation: 0 (real per yr), default (source, indicative — unverified)
info	Assumed default	Assumed Gas commodity price: Netherlands default (source, indicative — unverified)
info	Assumed default	Assumed Carbon price (Scope 1): Netherlands default (source, indicative — unverified)
info	Assumed default	Assumed WACC (real): 0.07 real, default (judgement, indicative — unverified)
info	Assumed default	Assumed Evaluation period: 20 yr, default (judgement, indicative — unverified)
info	Assumed default	Assumed UPS ride-through: 5 (min at full IT), default (judgement, indicative — unverified)
info	Assumed default	Assumed PPA type: hybrid, default (judgement, indicative — unverified)

Severity	Category	Flag
info	Design check	Design check at n = 50 (24.5 °C): peak PUE 1.292, design facility load 983.8 MW (+0.0 MW over the n = 20 sizing); not used for sizing
info	Assumed default	Technology, cost, carbon and other defaults: 550 of 574 registry records are indicative — unverified (PDF report section 2; user guide appendix)

#6 Grid + gas recips bridge -> retain as backup · Fuel cells top-up · retained-plant backup · Evaporative-adiabatic · PV 25 MWp · BESS for the shortfall · PPA hybrid 100 %

Severity	Category	Flag
major	Permitting	Aggregated combustion thermal input 2318 MWth >= 50 MWth (IED threshold). Screening flag, not a permitting determination.
minor	Resilience	Backup depends on gas network availability during a grid outage: plant that does not burn gas covers 0.0 of 916.3 MW critical load
minor	Grid	PPA limited by import capacity: 6,775,345 MWh (16.7 % of the 40,580,322 MWh undelivered) of contracted PPA volume over the 20-year evaluation period could not be delivered because the grid connection was full. Its settlement, a net cost of €112 m (present value €27.8 m), is reported as "PPA hedge (not counted)": it is not in the cost ledger, LCOE, NPV or score, because PPA settlement counts only on energy delivered over the grid connection.
minor	Grid	Battery for the shortfall cannot recharge from the grid headroom on the design day from month 80: it does not count as firm supply against the design load
minor	Permitting	Individual combustion units of 22.7 MWth (Gas reciprocating engines) are within 1–50 MWth (medium combustion plant rules). Screening flag, not a permitting determination.
minor	Weather	Weather: uploaded EPW file "NLD_Amsterdam062400_IWEC.epw" (AMSTERDAM, 1982–1999 (months from several years)), 189 km from the pin; sized to n = 20: 34.6 °C dry bulb and 24.5 °C wet bulb; check at n = 50: 36 °C and 24.5 °C, from the file's ASHRAE extremes (Climate Design Data 2009 ASHRAE Handbook): n = 20 and n = 50-year maximum dry bulb, and the extreme maximum wet bulb (no return period given for wet bulb). Warning: more than 50 km from the pin: check that its climate is representative
info	Grid	PPA volume above the load: 33,804,977 MWh (83.3 % of the 40,580,322 MWh undelivered) of contracted PPA volume over the 20-year evaluation period exceeded the load still to serve in its hour. Its settlement, a net cost of €112 m (present value €27.8 m), is reported as "PPA hedge (not counted)": it is not in the cost ledger, LCOE, NPV or score, because PPA settlement counts only on energy delivered over the grid connection.
info	Grid	Grid connection charge assumed €0 (the default, indicative — unverified); at the standard rate of €120k per MW (EHV) the 950 MW contracted would cost about €114.0 m
info	Permitting	Permitting outlook: Gas reciprocating engines (bridging; IED): uncertain; ready month 21, needed month 24, 3 months' float. Screening flag, not a permitting determination.
info	Fuel and utilities	Gas capacity and date assumed, not confirmed
info	Data	Result depends on C-tagged (indicative — unverified) parameters and synthetic profiles
info	Assumed default	Assumed Analysis start: 2027-01, default (judgement, indicative — unverified)
info	Assumed default	Assumed Average IT utilisation: 0.8 (fraction of installed IT), default (judgement, indicative — unverified)
info	Assumed default	Assumed Ramp: share at energisation: 0.25 (fraction of phase IT at energisation), default (judgement, indicative — unverified)
info	Assumed default	Assumed Load shape: daily, default (judgement, indicative — unverified)
info	Assumed default	Assumed Grid non-commodity adder: Netherlands default (source, indicative — unverified)
info	Assumed default	Assumed Electricity price escalation: 0 (real per yr), default (source, indicative — unverified)
info	Assumed default	Assumed Gas commodity price: Netherlands default (source, indicative — unverified)
info	Assumed default	Assumed Carbon price (Scope 1): Netherlands default (source, indicative — unverified)
info	Assumed default	Assumed WACC (real): 0.07 real, default (judgement, indicative — unverified)
info	Assumed default	Assumed Evaluation period: 20 yr, default (judgement, indicative — unverified)
info	Assumed default	Assumed UPS ride-through: 5 (min at full IT), default (judgement, indicative — unverified)
info	Assumed default	Assumed PPA type: hybrid, default (judgement, indicative — unverified)
info	Design check	Design check at n = 50 (24.5 °C): peak PUE 1.292, design facility load 983.8 MW (+0.0 MW over the n = 20 sizing); not used for sizing
info	Assumed default	Technology, cost, carbon and other defaults: 550 of 574 registry records are indicative — unverified (PDF report section 2; user guide appendix)

#9 Grid + gas recips bridge -> retain as backup · Fuel cells top-up · HVO N+1 · Evaporative-adiabatic · PV 25 MWp · BESS 2 h · PPA hybrid 100 %

Severity	Category	Flag
major	Permitting	Aggregated combustion thermal input 2653 MWth >= 50 MWth (IED threshold). Screening flag, not a permitting determination.
minor	Resilience	Backup depends on gas network availability during a grid outage: plant that does not burn gas covers 154.0 of 916.3 MW critical load
minor	Grid	PPA limited by import capacity: 6,775,345 MWh (16.7 % of the 40,580,322 MWh undelivered) of contracted PPA volume over the 20-year evaluation period could not be delivered because the grid connection was full. Its settlement, a net cost of €112 m (present value €27.8 m), is reported as "PPA hedge (not counted)": it is not in the cost ledger, LCOE, NPV or score, because PPA settlement counts only on energy delivered over the grid connection.
minor	Permitting	Individual combustion units of 22.7 MWth (Gas reciprocating engines), 7.2 MWth (HVO standby gensets) are within 1–50 MWth (medium combustion plant rules). Screening flag, not a permitting determination.
minor	Permitting	Permitting outlook: Gas reciprocating engines (bridging; IED): uncertain; ready month 21, needed month 24, 3 months' float; HVO standby gensets (backup; IED): likely; ready month 27, needed month 24, 3 months late. Screening flag, not a permitting determination.
minor	Weather	Weather: uploaded EPW file "NLD_Amsterdam062400_IWEC.epw" (AMSTERDAM, 1982–1999 (months from several years)), 189 km from the pin; sized to n = 20: 34.6 °C dry bulb and 24.5 °C wet bulb; check at n = 50: 36 °C and 24.5 °C, from the file's ASHRAE extremes (Climate Design Data 2009 ASHRAE Handbook): n = 20 and n = 50-year maximum dry bulb, and the extreme maximum wet bulb (no return period given for wet bulb). Warning: more than 50 km from the pin: check that its climate is representative
info	Grid	PPA volume above the load: 33,804,977 MWh (83.3 % of the 40,580,322 MWh undelivered) of contracted PPA volume over the 20-year evaluation period exceeded the load still to serve in its hour. Its settlement, a net cost of €112 m (present value €27.8 m), is reported as "PPA hedge (not counted)": it is not in the cost ledger, LCOE, NPV or score, because PPA settlement counts only on energy delivered over the grid connection.
info	Grid	Grid connection charge assumed €0 (the default, indicative — unverified); at the standard rate of €120k per MW (EHV) the 950 MW contracted would cost about €114.0 m
info	Fuel and utilities	Gas capacity and date assumed, not confirmed
info	Data	Result depends on C-tagged (indicative — unverified) parameters and synthetic profiles
info	Assumed default	Assumed Analysis start: 2027-01, default (judgement, indicative — unverified)
info	Assumed default	Assumed Average IT utilisation: 0.8 (fraction of installed IT), default (judgement, indicative — unverified)
info	Assumed default	Assumed Ramp: share at energisation: 0.25 (fraction of phase IT at energisation), default (judgement, indicative — unverified)
info	Assumed default	Assumed Load shape: daily, default (judgement, indicative — unverified)
info	Assumed default	Assumed Grid non-commodity adder: Netherlands default (source, indicative — unverified)
info	Assumed default	Assumed Electricity price escalation: 0 (real per yr), default (source, indicative — unverified)
info	Assumed default	Assumed Gas commodity price: Netherlands default (source, indicative — unverified)
info	Assumed default	Assumed Carbon price (Scope 1): Netherlands default (source, indicative — unverified)
info	Assumed default	Assumed WACC (real): 0.07 real, default (judgement, indicative — unverified)
info	Assumed default	Assumed Evaluation period: 20 yr, default (judgement, indicative — unverified)
info	Assumed default	Assumed Backup autonomy: 48 h, default (judgement, indicative — unverified)
info	Assumed default	Assumed Generator test hours: 6 (h/yr per unit), default (judgement, indicative — unverified)
info	Assumed default	Assumed UPS ride-through: 5 (min at full IT), default (judgement, indicative — unverified)
info	Assumed default	Assumed PPA type: hybrid, default (judgement, indicative — unverified)
info	Design check	Design check at n = 50 (24.5 °C): peak PUE 1.292, design facility load 983.8 MW (+0.0 MW over the n = 20 sizing); not used for sizing
info	Assumed default	Technology, cost, carbon and other defaults: 550 of 574 registry records are indicative — unverified (PDF report section 2; user guide appendix)

Run notes

- Compatibility rules: 64,980 impossible combinations not generated; 271,800 collapsed as n/a
- Values not generated by default or excluded: 19 choices
- Sampling above cap active: 2,000 of 52,023 combinations screened
- Site layout (Beta): this large campus was not laid out option by option in the run (layout work 38.9 million block2, limit 20 million), because site fit does not change the ranking. The Site tab lays out each option it shows. To lay out every option in the run, set "Site layout in the run" to every option
- 1,610 of 2,000 options serve their load in every hour; the other 390 rank below them (power adequacy first)

- IT utilisation assumption 80 % of installed IT (energy); sizing uses 100 %

9. Method

- Generation: every combination of the technology choices allowed by the compatibility rules, sampled above the generation cap with the archetypes always included.
- Sizing and dispatch: each option is sized from the design-day load and dispatched hour by hour (8,760 hours) for each year until its supply stops changing; later years repeat the last modelled year.
- Economics: one yearly cash-flow ledger per option gives the LCOE, the NPV and every cost breakdown (real terms, at the WACC).
- Carbon: location-based and market-based Scope 1 and 2, and hourly carbon-free energy (a PPA counts only against grid energy used in the same hour).
- Time-to-power: first power and full load from the programme of each technology (ordered from day one, with a six-month permitting allowance for plant that needs significant environmental permitting); the axis counts the months late against the phasing plan (first power against phase 1, full load against the last phase) plus half the power gap in months of design load. Rented bridging plant is hired a month before it is first needed.
- Ranking: power-adequate options first, then by the weighted score of the four axes, each scaled 0-100 across the run.
- Profiles: Synthetic profile (seeded), calibrated to indicative annual targets — not measured data.
- Run settings: generation cap 2000; grid handling "unmet"; run-hour handling "flag"; evaluation period 20 years; WACC 7.0 %.

Concept-stage screening only. Figures are indicative and must be verified with project-specific data and specialist advice before any design or investment decision. This tool is decision support, not financial or engineering advice.